

PRODUCT · ENGINEERING · STRATEGY

Elite Coach AI

Project Scope

MVP sprint through full product build — team structure, week-by-week delivery plan, technical decisions, and success criteria.

MVP TEAM SIZE

4
People

MVP TIMELINE

4
Weeks

FULL BUILD

3
Months

CORE FEATURES
(MVP)

3
Auth · Assessment · AI
Chat

PILOT TARGET

500+
Users by Month 3

0.1 What We're Building

Elite Coach AI is a web-based AI career mentorship platform targeting students, career switchers, and job seekers who lack access to professional guidance. The platform guides users from skill discovery through to job preparation — entirely through an AI-powered coaching interface. This scope covers the full build lifecycle: a focused MVP with a lean 4-person team, followed by a full product build over 3 months with an expanded team.

🕒 The core hypothesis the MVP must validate: **Users will repeatedly engage with an AI mentor to improve their career prospects.** Every feature decision must serve this one test. If users return and follow the roadmap, the MVP has done its job.

⚠️ The PRD lists five features for the MVP. This scope deliberately narrows the MVP to three, with the remaining two deferred to Month 1 of the full build. Five features with a 4-person team means none of them will be excellent. Three features that work flawlessly will win more users than five features that feel half-built.

0.2 MVP — What's In and What's Out

The MVP must prove the core engagement loop. That loop is: **Assessment → AI Roadmap → Coaching Conversation → User Returns.** Everything that does not serve that loop is out of scope for the MVP.

● In MVP ● Full Build Month 1 ● Post-MVP / Future

- **User Auth (Email + Google)** — Registration, login, basic profile. The entry point to everything else.
- **Skill Assessment (10-15 questions)** — Career interests, current skills, experience level. This feeds the AI with context. Without it, the AI gives generic answers.
- **AI Career Coach Chat** — The core product. Context-aware coaching powered by an LLM, with user's assessment data injected into every prompt. Chat history stored and included in follow-up sessions.
- **AI-Generated Learning Roadmap** — Generated immediately after assessment completion. Displayed as a step-by-step career path. Saved to the user's dashboard.
- **User Dashboard** — Shows name, career goal, roadmap progress checklist, and button to open AI chat. Simple. No clutter.
- **Resume Analyzer** — PDF/DOCX upload → AI feedback on structure, keywords, and improvements. Deferred to Month 1 of full build. Needs solid file parsing infrastructure.
- **Interview Practice** — AI-driven Q&A simulation with per-answer feedback. Deferred to Month 1 of full build. Needs its own UI state management and session logic.
- **AI Portfolio Builder** — Post-MVP. Phase 2 feature.
- **Job Matching / Internship Marketplace** — Post-MVP. Requires external integrations and significant data infrastructure.
- **Recruiter Integrations / Enterprise Partnerships** — Post-MVP. B2B sales motion, not MVP scope.

PHASE 1

MVP Sprint

4 Weeks
4 People

THE TEAM
FRONTEND DEVELOPER Web App — React / Next.js - Auth screens: sign up, login, Google OAuth - Onboarding + assessment flow (card-based, one question at a time) - AI chat interface — streaming responses, chat history display - Learning roadmap page — checklist with progress state - User dashboard — welcome, goal, roadmap, chat entry point - Responsive layout — must work on mobile browsers - Loading states and error handling on all AI-powered screens
BACKEND DEVELOPER API — Node.js + PostgreSQL - Auth API: JWT + Google OAuth, session management - User profile API: create, read, update - Assessment API: store responses, generate skill profile JSON - AI orchestration layer: build context-enriched prompts, call LLM API, stream responses - Chat history API: store and retrieve conversation threads - Roadmap API: save AI-generated roadmap, update progress - PostgreSQL schema: users, assessments, chats, roadmaps
UI/UX DESIGNER Product Design - Full design system: colors (Elite Global blue), typography, components - All MVP screens in Figma — auth, onboarding, assessment, chat, roadmap, dashboard - Chat UI design: message bubbles, typing indicators, response formatting - Roadmap visualization — clean step-by-step layout with progress indicators - Mobile-responsive layouts for all pages - Usability test with 3–5 real users before dev handoff (Week 1) - Component annotation and dev handoff specs
AI ENGINEER Intelligence Layer - Design system prompt: career coaching persona, tone, guardrails - Context injection: structure assessment data into every prompt - Roadmap generation prompt: structured JSON output → frontend renders it - Conversation memory: include prior chat history in each request - Token budget management: truncate history intelligently to stay within limits - LLM provider selection and API integration (OpenAI or Claude) - Test prompt quality across 10+ user types: student, switcher, job seeker
WEEK-BY-WEEK BREAKDOWN
WEEK 1 Foundation — Design, Auth, Infrastructure UI/UX Complete all MVP screen designs. Usability test with real users. Deliver Figma handoff by end of week.

- **BACKEND** Project setup: Node.js, Express, PostgreSQL. Auth API live (email + Google). Database schema created.
- **FRONTEND** Next.js scaffold. Design system integrated. Auth screens (sign up, login, Google) fully functional and connected to backend.
- **AI** LLM provider selected and API key configured. System prompt drafted and iterated. Test with 10 sample user profiles. Roadmap generation prompt producing structured output.

WEEK 2

Core Loop — Assessment + AI Chat + Roadmap

- **FRONTEND** Onboarding flow. Assessment questionnaire (card-by-card). AI chat interface with streaming. Roadmap page rendering from API response.
- **BACKEND** Assessment API. AI orchestration layer — receives assessment data, builds enriched prompt, calls LLM, streams response. Roadmap save/retrieve. Chat history store.
- **AI** Context injection tested end-to-end: assessment data → prompt → coaching response. Verify roadmap output is consistent and structured. Token management logic implemented.
- **UI/UX** Review implemented screens against Figma. Flag any deviations. Begin iterating on chat UI based on real AI responses.

WEEK 3

Dashboard, Polish & Integration

- **FRONTEND** User dashboard complete. Roadmap progress checklist with state persistence. Full user journey tested: sign up → assessment → roadmap → chat → return visit.
- **BACKEND** Progress tracking API. Roadmap step completion endpoint. Error handling and rate limit management on AI calls. Environment variables and secrets management.
- **AI** Returning user prompt: include roadmap + previous chat context. Test multi-session coherence — does the AI remember what it told the user last time? Tune response quality.
- **ALL** End-to-end integration test: complete user journey from zero to returning visit. Bug fixes. Performance check — chat response time under 3 seconds.

WEEK 4

QA, Deployment & Pilot Launch

- **ALL** Structured QA: test 10 complete user journeys across 3 personas (student, career switcher, job seeker). Log every AI response quality issue.
- **BACKEND** Deploy to production (Vercel + Railway or Render). Environment config. API rate limiting. Basic logging and error monitoring (Sentry).
- **FRONTEND** Mobile responsiveness audit. Loading state polish. Empty state designs (first-time user, no chat history, etc.).
- **AI** Final prompt tuning based on QA findings. Document the system prompt and context structure for the full build team. Identify failure modes observed in testing.
- **ALL** Soft launch to 20-30 pilot users. Read every chat transcript. Collect feedback. Prepare full build kickoff brief.

WHAT SHIPS AT END OF WEEK 4

• **Working web app** — Deployed, accessible via browser on desktop and mobile

• **Email + Google authentication** — Secure sessions, password reset flow

• **Skill assessment** — 10-15 questions, responses stored, skill profile generated

• **AI coaching chat** — Context-aware, persona-consistent, with conversation memory across sessions

• **Personalized learning roadmap** — AI-generated, saved to user's profile, rendered as interactive checklist

• **User dashboard** — Welcome, career goal, roadmap progress, chat access

• **System prompt documentation** — Written spec of how the AI is instructed and how context is built, ready for full build team

MVP SUCCESS CRITERIA — GATE TO FULL BUILD

METRIC

TARGET

HOW MEASURED

Assessment completion rate	≥ 70% of sign-ups	Analytics event tracking
Roadmap generation success rate	100% — no broken outputs	Backend logs + manual review
AI chat sessions per user (Week 1)	≥ 2 sessions per user	Chat session analytics
Day 7 retention	≥ 30% of pilot users return	Login analytics
Chat response time	First token under 3 seconds	Backend latency logs
AI response quality (manual review)	Founder rates ≥ 80% of responses as "useful"	Manual transcript review
App crash / error rate	< 2% of sessions	Sentry error logs
User satisfaction (pilot survey)	≥ 4.0 / 5.0	Post-session survey

❌ NOT BUILDING IN MVP

- × Resume upload and analysis — deferred to Full Build Month 1
- × Interview practice simulator — deferred to Full Build Month 1
- × Job matching or recruiter integrations — post-MVP
- × Mobile native app (iOS/Android) — web is sufficient for MVP validation
- × Payment / subscription system — validate free before monetising
- × AI-generated learning courses — post-MVP
- × Admin dashboard or analytics panel — log to Sentry/Mixpanel directly for now

⚠️ **The most important technical decision in the MVP:** How user context is structured and passed to the LLM. The AI engineer must solve this in Week 1 — not Week 3. Generic AI responses are a product-killer. If the coach sounds like it does not know who it is talking to, users will not come back.

04 Phase 2 — Full Product Build

PHASE 2

Full Product Build

3 Months

Expanded Team

Phase 2 starts after MVP success criteria are met and pilot feedback is synthesised. The original 4-person team leads, expanded with new hires. The goal is to add resume and interview features, improve AI quality, add monetisation, and grow to 500+ active users.

RECOMMENDED EXPANDED TEAM

CARRY-OVER

Frontend Developer (Lead)

CARRY-OVER

Backend Developer (Lead)

CARRY-OVER

UI/UX Designer (Lead)

CARRY-OVER

AI Engineer (Lead)

NEW HIRE

2nd Frontend Developer

NEW HIRE

QA / Testing Engineer

NEW HIRE

Growth / Product Analyst

NEW HIRE

2nd Backend / DevOps

ADVISORY

Career Coach (Domain Expert)

MONTH-BY-MONTH PLAN

MONTH 1 — COMPLETE THE PRODUCT

- Resume analyzer: PDF/DOCX upload, text extraction, AI feedback (structure, keywords, improvements)
- Interview practice: AI asks questions, user types answers, AI scores and coaches each response
- Roadmap editing: allow users to customise their AI-generated roadmap
- Notification system: email reminders to return and continue the roadmap
- Basic analytics dashboard for internal team (user counts, session stats)
- Fix all MVP issues identified in pilot feedback

MONTH 2 — QUALITY & RETENTION

- AI quality improvement: refine prompts using real conversation transcripts from Month 1
- Progress tracking: mark roadmap steps complete, show % progress visually
- Weekly check-in feature: AI proactively asks "what did you learn this week?"
- Subscription / payment integration (Paystack for Nigeria, Stripe for global)
- Free vs paid tier definition and feature gating
- Mobile web optimisation: smooth experience on all screen sizes
- Domain expert review: career coach reviews AI roadmap outputs for accuracy

MONTH 3 — GROWTH & SCALE

- User referral system: "invite a friend, get 1 month free"
- Career path library: pre-built starter roadmaps for top 20 career tracks
- Resource recommendations: AI links users to specific free courses, tutorials
- Admin dashboard: user management, churn alerts, AI usage costs
- SEO-optimised marketing pages for organic acquisition
- University / bootcamp partnership pipeline (B2B sales preparation)
- Public launch — Product Hunt, tech communities, social push

PHASE 2 SUCCESS CRITERIA

METRIC	TARGET
Total registered users	500+ by end of Month 3
Monthly active users (MAU)	≥ 40% of registered users active each month
Day 30 retention	≥ 25% of new users still active after 30 days
Resume analyses completed	≥ 200 by end of Month 1
Interview sessions completed	≥ 150 by end of Month 1

Paying users (after Month 2)	≥ 50 paying subscribers
Average sessions per active user/week	≥ 3 sessions per week
NPS score	≥ 40

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Recommended Tech Stack

FRONTEND - Next.js 14 (App Router) — SSR for SEO, fast page loads - Tailwind CSS — utility-first, fast to build - Vercel deployment — instant deploys, free tier sufficient for MVP - React Query — server state management for chat and roadmap - Streaming support for AI chat responses (EventSource / fetch streams)
BACKEND - Node.js + Express — aligns with PRD, familiar ecosystem - PostgreSQL — user accounts, assessments, chat history, roadmaps - Prisma ORM — type-safe database access - Railway or Render — easy managed PostgreSQL + API hosting - JWT authentication + Google OAuth (Passport.js)
AI LAYER - Primary: Claude (Anthropic API) — better instruction-following for structured coaching - Fallback: OpenAI GPT-4o — widely documented, large community - Streaming responses via SSE or WebSocket - Prompt templates stored as versioned config (not hardcoded in logic) - Token usage tracking per user (cost management from Day 1)
TOOLING & OPS - Sentry — error monitoring from Week 3 - Mixpanel or PostHog — user analytics, funnel tracking - GitHub + GitHub Actions — CI/CD pipeline - Resend or SendGrid — transactional email - Cloudinary or S3 — file uploads (resume files in Full Build)

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Key Risks

RISK	LEVEL	MITIGATION
AI gives generic, unhelpful responses	High	AI engineer focuses Week 1 entirely on prompt quality. Founder reads every conversation in the pilot week. Iterate the system prompt aggressively before Week 3.
Users drop off after first session and don't return	High	The roadmap must feel personal, not generic. Email re-engagement from Day 3. Week 7 check-in prompt built into the AI session itself. Track Day 3 and Day 7 retention obsessively.
4 weeks is tight for	Medium	Scope is deliberately 3 core features not 5. Week 4 is QA and deployment only —

4 features + deployment		no new feature work. Designer delivers all screens by end of Week 1 so dev never waits on design.
LLM API costs escalate unexpectedly	Medium	Token budget per conversation enforced from Day 1. Track cost per user in the backend. Set a hard monthly spend cap during MVP. Compress chat history intelligently (summarise old messages).
Assessment questions don't give AI enough context	Medium	AI engineer designs the assessment questions, not just the responses. The questions must extract actionable signal: specific skills known, target role, timeline, biggest blocker. Test with 5 real people before launch.
Chat history grows too long, context window fills up	Low-Medium	Implement conversation summarisation after 20 messages: compress old messages into a running summary. Include summary + last 5 messages in each new prompt. Solved in Week 2 backend work.
